YTU – Faculty of Chemical and Metallurgical Engineering, Questions and Answers Sheet		NOTE CHART		
		Q1-Q4	Q5-Q7	Total
Name and Surname				
Student Number				
Course Name	Theory of Linear Programming (2nd midterm)	Duration		Date: 21/12/2023
Course Instructor Name and Surname	Dr. Gökhan GÖKSU		Signature	

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Q1. 5p. The right simplex tableau indicates:
a. a unique optimal solution
b. alternate optimal solutions
c. an infeasible solution
d. a feasible solution but not optimal
e. a feasible but unbounded

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	rhs
max Z	1	0	0	0	2	0	1	5	-10
	3	0	2	1	1	0	0	0	6
	-1	0	-2	0	-1	1	1	1	1
	4	1	1	0	3	0	2	-3	4

Q2. 5p. The right simplex tableau indicates:
a. a unique optimal solution.
b. alternate optimal solutions
c. an infeasible solution
d. optimal solution
e. a feasible but unbounded solution

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	rhs
max Z	1	0	-3	0	2	0	-1	5	-10
	3	0	0	1	1	0	0	0	6
	-1	0	-2	0	-1	1	1	1	1
	4	1	-4	0	3	0	2	-3	4

Q3. 15p. The first (starting) and the second tableau of a linear programming problem are given on the right. Find:

i. a+c+e=____

iii. g + i =____

iv. f + k + j =____

Basic	x_1	x_2	x_3	x_4	x_5	rhs
Z	a	-1	3	0	0	0
	b	c	d	1	0	6
	-1	2	e	0	1	1
Basic	x_1	x_2	x_3	x_4	x_5	rhs
Z	0	-1/3	j	k	m	2
	g	2/3	2/3	1/3	0	f
	h	i	-1/3	1/3	1	3

CEVAP:

Taban	x_1	x_2	x_3	x_4	x_5	rhs
Z	-1 a	-1	3	0	0	0
$x_4 \rightarrow$	b 3	c 2	d 2	1	0	6
x_5	-1	2	e -1	0	1	1
Taban	x_1	x_2	x_3	x_4	x_5	rhs
Z	0	-1/3	j $3+\frac{2}{3}=\frac{11}{3}$	k 1/3	m 0	2
x_1	g 1	2/3	2/3	1/3	0	f 2
x_5	h 0	i 8/3	-1/3	1/3	1	3

Q4. Consider the given problem and its intermediate iteration such that $x_B = [x_3 \quad x_2]^T$, $x_N = [x_1 \quad x_4]^T$. Selecting x_1 as the entering variable, determine the following for the next iteration:

$$\min Z = -2x_1 - 3x_2$$
$$\text{s.t.} \quad x_1 + x_2 + x_3 = 6$$
$$x_1 + 3x_2 + x_4 = 12$$
$$x_1, x_2, x_3, x_4 \geq 0$$

i. 5p. B of linearly independent columns of the original coefficient matrix.

a. $\begin{bmatrix}1 & 1\\0 & 3\end{bmatrix}$ b. $\begin{bmatrix}1 & 1\\1 & 3\end{bmatrix}$ c. $\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ d. $\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$ e. $\begin{bmatrix}1 & 0\\3 & 1\end{bmatrix}$

ii. 10p. z-row coefficients of basic variables x_B (cost information):

a. $[-3/2 \quad -1/2]^T$ b. $[-1/2 \quad -3/2]^T$ c. $[0 \quad 0]^T$ d. $[1/2 \quad 3/2]^T$ e. $[3/2 \quad 1/2]^T$

iii. 5p. the rhs vector:

a. $\begin{bmatrix}0\\3\end{bmatrix}$ b. $\begin{bmatrix}3\\3\end{bmatrix}$ c. $\begin{bmatrix}3\\0\end{bmatrix}$ d. $\begin{bmatrix}2\\4\end{bmatrix}$ e. $\begin{bmatrix}4\\2\end{bmatrix}$

$b = \begin{bmatrix}6\\12\end{bmatrix}$

i) $X_B = [x_3 \quad x_2]^T$, $X_N = [x_1 \quad x_4]^T$ $A = \begin{array}{c} \\ x_3 \\ x_2 \end{array}\begin{array}{c} x_1 \quad x_2 \quad x_3 \quad x_4 \\ \begin{bmatrix}1 & 1 & 1 & 0\\1 & 3 & 0 & 1\end{bmatrix}\end{array}\begin{array}{c} \\ 6 \\ 12 \end{array}$

$B = \begin{bmatrix}1 & 1\\0 & 3\end{bmatrix}$ $N = \begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}$, $c_B = [0 \; -3]^T$

$c_N = [-2 \; 0]^T$

$B^{-1} = \begin{bmatrix}1 & -1/3\\0 & 1/3\end{bmatrix}$ $c_B^T B^{-1} N - c_N^T = [1 \quad -1]^T$ (current cost information)

$[0 \; -3]\begin{bmatrix}1/3 & -1/3\\1/3 & 1/3\end{bmatrix}$

x_1 should be the entering variable

$\min\left\{\frac{6}{1}, \frac{12}{1}\right\} = 6 \rightarrow x_3$ should leave the basis

Therefore B_{new} should be $\begin{array}{c} x_1 \quad x_2 \\ \begin{bmatrix}1 & 1\\1 & 3\end{bmatrix}\end{array}$ ✓ B

a.

X

For the next iteration

$X_B = [x_1 \quad x_2]^T$ $X_N = [x_3 \quad x_4]^T$

$B = \begin{array}{c} x_1 \quad x_2 \\ \begin{bmatrix}1 & 1\\1 & 3\end{bmatrix}\end{array}$, $N = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$, $B^{-1} = \begin{bmatrix}3/2 & -1/2\\-1/2 & 1/2\end{bmatrix}$

$c_B = [-2 \; -3]^T$, $c_N = [0 \quad 0]^T$

b

$c_B^T B^{-1} N - c_N^T = [-3/2 \; -1/2]$ ✓ A

$RHS = B^{-1} b = \begin{bmatrix}3/2 & -1/2\\-1/2 & 1/2\end{bmatrix}\begin{bmatrix}6\\12\end{bmatrix} = \begin{bmatrix}3\\3\end{bmatrix}$

c

Q5) 5p. Consider a primal problem and its dual. Which of the below statement(s) is(are) correct?
i. Inequality constraints in the primal appear as a restriction on the variables in the dual.
ii. Nonnegative variables in the primal appear as nonpositive variables in the dual.
iii. Equality constraints in the primal appear as unrestricted variables in the dual.

A. Only i B. Only ii C. Only iii D. i and iii E. ii and iii

Q6. 15p. Which of the following is the dual of the given LP problem?	$\max Z = 4x_1 + 3x_2$ $s.t. \quad x_1 + 5x_2 \geq 8$ $2x_1 + x_2 = 6$ $6x_1 + x_2 \leq 2$ $x_1 \text{ free}, x_2 \geq 0$

a.
$$\begin{aligned} &\min W = 8y_1 + 6y_2 + 2y_3 \\ &s.t. \quad y_1 + 2y_2 + 6y_3 \geq 4 \\ &\quad 5y_1 + y_2 + y_3 \geq 3 \\ &\quad y_1, y_2, y_3 \text{ free} \end{aligned}$$

b.
$$\begin{aligned} &\min W = 8y_1 + 6y_2 + 2y_3 \\ &s.t. \quad y_1 + 2y_2 + 6y_3 = 4 \\ &\quad 5y_1 + y_2 + y_3 \geq 3 \\ &\quad y_1 \leq 0, y_2 \text{ free}, y_3 \geq 0 \end{aligned}$$

c.
$$\begin{aligned} &\min W = 8y_1 + 6y_2 + 2y_3 \\ &s.t. \quad y_1 + 2y_2 + 6y_3 = 4 \\ &\quad 5y_1 + y_2 + y_3 \leq 3 \\ &\quad y_1, y_2, y_3 \geq 0 \end{aligned}$$

Q7.Consider the following linear programming problem and its optimal table where S_1, S_2 and S_3 are the slack variables of the first, the second and the third constraint, respectively.

$$\begin{aligned} &\max Z = 4x_1 + 3x_2 \\ &s.t. \quad 2x_1 + 3x_2 \leq 1200 \\ &\quad 2x_1 + x_2 \leq 1000 \\ &\quad 4x_2 \leq 800 \\ &\quad x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Basic	x_1	x_2	S_1	S_2	S_3	rhs
Z	0	0	1/2	3/2	0	2100
x_2	0	1	1/2	-1/2	0	100
x_1	1	0	-1/4	3/4	0	450
S_3	0	0	-2	2	1	400

i. 5p. What is the dual price of the first constraint?

a. 1/2 b. 3/2 c. 0 d. -1/2 e. -3/2

ii. 10 p. Find the optimality range of the first constraint.

a. $[800, 1300]$
b. $[950, 1350]$
c. $[1200, 2000]$
d. $[1000, 1400]$

$$\begin{aligned} &\max Z = 4x_1 + 3x_2 \\ &s.t. \quad 2x_1 + 3x_2 \leq 1200 \\ &\quad 2x_1 + x_2 \leq 1000 \\ &\quad 4x_2 \leq 800 \\ &\quad x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Basic	x_1	x_2	S_1	S_2	S_3	rhs
Z	0	0	1/2	3/2	0	2100
X2	0	1	1/2	-1/2	0	100
X1	1	0	-1/4	3/4	0	450
S3	0	0	-2	2	1	400

ii. 10 p. Find the optimality range of the first constraint.

$D_2 = D_3 = 0$
D_1 ?

$X_2 = 100 + \frac{1}{2}D_1 - \frac{1}{2}D_2$
$X_1 = 450 - \frac{1}{4}D_1 + \frac{3}{4}D_2$
$S_3 = 400 - 2D_1 + 2D_2 + D_3$

$100 + \frac{D_1}{2} \geq 0, \quad D_1 \geq -200$
$450 - \frac{D_1}{4} \geq 0, \quad D_1 \leq 1800$
$400 - 2D_1 \geq 0, \quad D_1 \leq 200$

$-200 \leq D_1 \leq 200$

$\boxed{1000 \leq RHS_1 \leq 1400}$

iii.10p. If the rhs of constraints are changed into 1100, 1100, and, 700 units, respectively, then (x_1,x_2) in the new optimum solution is:

a. $(550, 0)$

b. $(500, 100)$

c. $(400, 100)$

d. $(350, 200)$

e. Post-optimality analysis is required.

$RHS_{S_1} = 1100$, $RHS_{S_2} = 1100$, $RHS_{S_3} = 700$

$D_1 = -100$, $D_2 = +100$, $D_3 = -100$.

$x_1 = 550$
$x_2 = 0$
$S_3 = 700$

$z^* = 2200$.

iv. 10p. Find the feasibility range for the objective function coefficient of x_1?

a. $\left[\frac{3}{2}, \frac{11}{2}\right]$ b. $\left[\frac{3}{2}, 5\right]$ c. $[2, 6]$ d. $[4, 5]$

e. Post-optimality analysis is required.

$\max Z = 4x_1 + 3x_2$

s.t. $2x_1 + 3x_2 \le 1200$

$2x_1 + x_2 \le 1000$

$4x_2 \le 800$

$x_1 \ge 0, x_2 \ge 0$

	Basic	x_1	x_2	S_1	S_2	S_3	rhs
		λ	0	0	0	0	
1	Z	0	0	1/2	3/2	0	2100
0	X2	0	1	1/2	-1/2	0	100
λ	X1	1	0	-1/4	3/4	0	450
0	S3	0	0	-2	2	1	400

RC of $x_1 = 0$
RC of $x_2 = 0$
RC of $S_1 = \frac{1}{2} - \frac{1}{4}\lambda \ge 0$ $\quad \frac{1}{2} \ge \frac{\lambda}{4}$ $\quad 4 \ge 2\lambda$ $\quad 2 \ge \lambda$

RC of $S_2 = \frac{3}{2} + \frac{3}{4}\lambda \ge 0$ $\quad \frac{\cancel{3}}{4}\lambda \ge -\frac{\cancel{3}}{2}$ $\quad \boxed{\lambda \ge -2}$

R.C. of $S_3 = 0$

$\lambda \in [-2, 2]$.

Original cost $= 4$

$\boxed{2 \le c_1 \le 6}$